L-T-1313A
January 6, 1977
SUPERSEDING
Int. Fed. Spec. L-T-001313 (DSA-DM)
May 22, 1967

FEDERAL SPECIFICATION

TUBING, STERILIZATION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers nylon, transparent, lay-flat tubing suitable for fabrication as a container for surgical packs to be sterilized by steam sterilization after sealing, and to maintain sterility after autoclaving.

1.2 Classification. Nylon tubing shall be flat and shall be of the following sizes, as specified below (see 6.2). All sizes shall be 500 feet in length.

Size I	6	inches wide
Size II	12	" "
Size III	18	" "
Size IV	2	" "
Size V	4	" "
Size VI	8	" "

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D-789 - Nylon Injection Molding and Extrusion Materials.
D-792 - Specific Gravity and Density of Plastics by Displacement.
D-882 - Tensile Properties of Thin Plastic Sheeting.
D-1434 - Gas Transmission Rate of Plastic Film and Sheeting.
D-2863 - Flammability of Plastics Using the Oxygen Index Method.
E-96 - Water Vapor Transmission of Materials in Sheet Form.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Material.

3.2.1 Nylon resin. The material used in the tubing film shall be Nylon 6, non-plasticized, virgin, first grade resins, which when produced into film shall be readily sealable by a sealing iron. The resin shall have the following properties when tested as specified in 4.4.1.

Melting Point 213° C to 220° C (415° F to 428° F).
Specific Gravity 1.13 to 1.14 at 4° C (39° F).

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3.2.2 Finished film. The finished film shall be transparent, .002 inches thick ± 10 %, and shall have the following properties, when tested as specified in 4.4.2.

Tensile Strength	9500 - 12,500 pounds per square inch.
Elongation	300 - 350 percent.
Flammability	No specimen, out of 5 tested, shall burn more than 3 minutes nor more than 50 mm at an oxygen level of 31%.

3.2.3 Tubing.

3.2.3.1 Autoclaving. When autoclaved as specified in 4.4.3.1, the tubing shall show no breaks nor discoloration and the minimum percentages of original tubing tensile strength given in table I, and original elongation given in table II, shall be retained.

TABLE I. Minimum tensile strength (percent of original)

Autoclaving Time	Machine Direction	Transverse Direction
1 hr.	100	80
2 hrs.	100	80
3 hrs.	100	80
4 hrs.	90	80
5 hrs.	90	70
6 hrs.	90	70

TABLE II. Minimum elongation (percent of original)

Autoclaving Time	Machine Direction	Transverse Direction
1 hr.	80	80
2 hrs.	70	70
3 hrs.	70	60
4 hrs.	60	50
5 hrs.	50	40
6 hrs.	40	30

3.2.3.2 Permeability. The tubing shall be permeable to gas and water vapor as indicated below.

3.2.3.2.1 Gas permeability. Gas permeability, measured in cubic centimeters/100 square inches/24 hours/.001 inch of thickness/1 standard atmosphere pressure differential, at a temperature of 25° C, (77° F) when tested as specified in 4.4.3.2, shall be:

Oxygen	2.6 ± 10%
Nitrogen	0.9 ± 10%
Carbon Dioxide	4.7 ± 10%

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3.2.3.2.2 Water vapor transmission rate. Water vapor transmission rate shall be 19.5 grams + 10% /100 square inches/24 hours/.001 inch of thickness at 38° C (100° F) and 90% relative humidity when tested as specified in 4.4.3.2.

3.2.3.3 Chemical resistance. The tubing shall be inert to oils, fats, waxes and grease and shall not be adversely affected by Butyl alcohol, Butyl Chlorobenzene, Acetone, Methyethyl Ketone, Toluene, or Chlorobenzene (Mineral acids, formic acid and phenol will attack the film, however.) The tubing shall not be used for packaging water-based solutions or products which are highly sensitive to moisture.

3.2.3.4 Sterilization. When tested as specified in 4.4.4, the contents of the packs tested immediately after sterilization, and those tested 30 days later, shall show no evidence of bacterial or mold growth.

3.3. Design and construction.

3.3.1 Tubing. Shall be extruded, flat, transparent, nylon-film tubing. The tubing shall be free of pinholes, wrinkles, blisters, streaks, and crow-feet, dirt and foreign matter. The tubing shall be wound firmly on a paperboard core. Each roll shall consist of at least 500 linear feet of tubing. Any one roll shall contain no more than one break in continuity. The width of the tubing shall be as indicated in table III.

TABLE III. Tubing width

Size	Width (inches)
I	6 ± 1/4
II	12 ± 1/4
III	18 ± 1/4
IV	2 ± 1/4
V	4 ± 1/4
VI	8 ± 1/4

3.3.1.1 Sealability. The tubing shall be capable of being sealed with the use of a suitable sealing iron (see 6.5). The seal shall remain intact after being tested as specified in 4.4.3.1.

3.4 Workmanship. Workmanship shall be first class throughout. The tubing and its container shall be free from defects which detract from their appearance or may impair their serviceability. The rolled tubing, of each size, shall have the edges aligned within 3/8 inch at each end on any radius. Irregularities in winding the tubing on the core which result in the first few turns (not exceeding 10 feet) extending beyond the reel shall not be cause for rejection. These first few turns extending beyond the core shall be trimmed, and an additional 10 feet of tubing shall be included in each roll to compensate for the trimmed length.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the tubing when a first article sample is required (see 3.1). This inspection shall include the examination specified in 4.3 and tests specified in 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination specified in 4.3 and tests specified in 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.1.4 Certificate of compliance. A certificate of compliance, supplied by the manufacturer of the tubing, may be furnished by the contractor in lieu of actual performance of the test specified in 4.4.4. The Government reserves the right to verify compliance with the requirement by testing a sample, representative of the tubing being supplied, from each lot.

4.2 Sampling.

4.2.1 For examination. Sampling for the examinations specified in 4.3 and 4.3.2 shall be conducted in accordance with MIL-STD-105, and table IV. The unit of product for sampling purposes shall be one roll.

TABLE IV. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	S-2	1.0
Minor defects	S-2	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table V. The acceptance number shall be zero defects for tests specified in table V.

TABLE V. Sampling for tests

Component (Unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Nylon resin	Melting point	3.2.1	4.4.1	1/
	Specific gravity	3.2.1	4.4.1	1/
Nylon film	Tensile strength	3.2.2	4.4.2	S-1
	Elongation	3.2.2	4.4.2	S-1
	Flammability	3.2.2	4.4.2	1/
Tubing (roll)	Autoclaving	3.2.3.1	4.4.3.1	S-1
	Gas permeability	3.2.3.2.1	4.4.3.2	S-1
	Water-vapor transmission	3.2.3.2.2	4.4.3.2	S-1
	Sterilization	3.2.3.4	4.4.4	1/

1/ One sample from each lot. An inspection lot for inspection of tubing shall consist of all tubing produced under essentially the same conditions at the same time, using the same lot of raw material, and offered for inspection at one time.

4.3 Examination. The tubing shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table VI. Examination shall not be restricted to the classified possible defects listed below.

TABLE VI. Classification of defects

Catagories	Defects
Major:	
101	Tubing has pinholes.
102	Tubing wrinkled.
103	Tubing has streaks, crow-feet, or blisters.
104	Tubing not firmly and evenly rolled on core.
105	More than one break in continuity in any one roll.
Minor:	
201	Finished roll not round.
202	Paperboard core extends more than 1/4 inch beyond roll at either end.
203	Dirt or foreign matter on tubing.
204	Tubing does not unroll freely from box.
205	Flap top of box missing or not free-moving.

4.3.2 Dimensional examination. The tubing shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Nylon resin. Testing of nylon resin shall be in accordance with the ASTM Standard Methods of Test listed in table VII.

TABLE VII. Testing of nylon resin

Characteristic	Test Method
Melting point	ASTM D-789
Specific gravity	ASTM D-792

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4.4.2 Nylon film. Testing of nylon film shall be in accordance with the ASTM Standard Methods of Test listed in table VIII.

TABLE VIII. Testing of nylon film

Characteristic	Test Method
Tensile strength	ASTM D-882
Elongation	ASTM D-882
Flammability	ASTM D-2863

4.4.3 Tubing.

4.4.3.1 Autoclaving. A suitable quantity of 4 x 4 gauze shall be prepared in cloth wrappers and overpacked with the nylon tubing. Push the air out of the pack and heat seal. Place the sealed pack in a general purpose autoclave and subject it to a normal sterilizing cycle with steam at 136° C. (277° F.) and pulling 20 inches of vacuum. The test shall be repeated for a total of 3 complete cycles.

4.4.3.2 Permeability. The tubing shall be tested for permeability to gas and water vapor in accordance with the ASTM Standard Methods of Test listed in table IX.

TABLE IX. Testing of tubing for permeability

Characteristic	Test Method
Gas permeability	ASTM D-1434
Water vapor transmission rate	ASTM E-96, Procedure E

4.4.4 Sterilization test. Each of twenty samples of the tubing shall be packed with a suitable quantity of 4 x 4 gauze, prepared in a cloth wrapper, and a spore strip impregnated with Bacillus Stearothermophilus. The air shall be pushed out the pack and they shall be heat sealed. The sealed packs shall be placed in a general purpose autoclave and subjected to a normal sterilizing cycle with steam at 136° C (277° F) and pulling 20 inches of vacuum. Immediately after sterilization, the contents of ten packs shall be tested for sterility as specified below. The remaining ten packs shall be exposed to the atmosphere at ambient temperature (18° C to 27° C) (65° F to 80° F) for 30 days and the contents shall then be tested for sterility. To test for sterility, aseptically transfer five of the gauze packs and all ten of the spore strips to culture tubes containing Fluid Soybean Casein Digest. Incubate the gauze packs at 20° C to 25° C and incubate the spore strips at 55° C to 65° C for not less than seven days. Transfer the five remaining gauze packs to culture tubes containing Thioglycollate Medium and incubate at 30° C to 35° C for not less than seven days. The quantity and preparation of the media shall be as described under Sterility Tests in the U.S. Pharmacopeia.

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4.5 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table X. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE X. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.5.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Five hundred feet of tubing of the size specified shall be neatly wound on a chipboard or paperboard core having an internal diameter of one inch, plus 1/8 inch, minus 0 inch and a wall thickness of not less than 1/32 inch. The core shall not extend more than 1/4 inch beyond the tubing. The tubing shall be further packaged in a box conforming to PPP-B-566. The box shall be designed with a flap top so that the tubing may be unrolled directly from box without opening the box when the tubing is passed between the flap and the adjacent side wall of the box.

5.1.2 Level C. Five hundred feet of tubing shall be neatly wound on a core and packaged in a dispenser type box that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

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5.2.1 Level A. Six unit packages containing size I, four unit packages containing size II, two unit packages containing size III, twenty-four unit packages containing size IV, twelve unit packages containing size V, or four unit packages containing size VI shall be packaged in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, waterproofing and bonding shall be as specified in the appendix of the box specification.

5.2.2 Level B. Tubing shall be packaged as specified in 5.2.1 except that the exterior container shall conform to PPP-B-636, class domestic and waterproofing and bonding shall not be required.

5.2.3 Level C. The packaged tubing shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tubing is intended for use in packaging medical supplies for steam autoclaving in general purpose sterilizers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Size required (see 1.2).
d. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

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6.4 This specification covers the following item(s) appearing in the 'ederal Supply Catalog:

National Stock Number	Item Identification	Specification Designation Size
6530-00-104-7530	TUBING, STERILIZATION, Nylon 6 inches Wide Flat, 500 feet	I
6530-00-104-7531	TUBING, STERILIZATION, Nylon 12 inches Wide Flat, 500 feet	II
6530-00-104-7532	TUBING, STERILIZATION, Nylon 18 inches Wide Flat, 500 feet	III
6530-00-104-7533	TUBING, STERILIZATION, Nylon 2 inches Wide Flat, 500 feet	IV
6530-00-104-7534	TUBING, STERILIZATION, Nylon 4 inches Wide Flat, 500 feet	V
6530-00-105-6794	TUBING, STERILIZATION, Nylon 8 inches Wide Flat, 500 feet	VI

6.5 Sealing irons. Sealing irons which meet the requirements of 3.3.1.1 re available under NSN 3540-00-299-9811. Qualified manufacturers include alad Electric Heating, Inc., 1600 Widley St., Tarrytown, NY 10591 and ertrod Corp., 2039 Utica Ave., Brooklyn, NY, 11234.

6.6 Cross reference. The correlation between the sizes of tubing used n this specification and the previous size designations used in Interim 'ederal Specification L-T-001313, are indicated in table XI.

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TABLE XI Correlation of tubing sizes

Size Designation	Former size (L-T-001313)	Current size (L-T-1313A)
I	2 inches wide, 100 feet long	6 inches wide, 500 feet l
II	4 inches wide, 500 feet long	12 inches wide, 500 feet l
III	6 inches wide, 100 feet long	18 inches wide, 500 feet l
IV	8 inches wide, 500 feet long	2 inches wide, 500 feet l
V	12 inches wide, 100 feet long	4 inches wide, 500 feet l
VI	18 inches wide, 500 feet long	8 inches wide, 500 feet l

MILITARY CUSTODIANS:
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NAME OF ORGANIZATION AND ADDRESS

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